AML-3406

AL and ML Capstone Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 9
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	**Bansil Patel**	**Harsh Mohile**	**Meet Patel**	**Rachit Bhatt**
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Completed Docker Installation and successfully created containers for the project. Explored the use of Docker Compose for handling multiple containers.

Bansil Patel & Meet Patel: Conducted performance evaluations of different models and began the process of tuning hyperparameters on the selected model.

Harsh Mohile & Rachit Bhatt: Showcased the setup and execution of CI/CD pipelines using Github Actions and Docker. Worked on optimizing Github Actions workflows for automatic deployment.

2.2 Your team's tasks

- **Model Application:** Progressed in evaluating alternative models and started the hyperparameter tuning process based on the model performance comparison.
- **CI/Cd Pipeline Demonstration:** Successfully demonstrated how Github Actions can be used to automate Dcker deployments, streamlining the CI/CD process for the project.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Set up Docker containers and explored Docker Compose for multi-container management.

Bansil Patel & Meet Patel: Evaluated model performance (accuracy, speed) and started hyperparameter tuning on the selected model.

Harsh Mohile & Rachit Bhatt: Demonstrated automated deployment using Github Actions and Docker, optimizing windows.

3.2 Your team's progress

- Configured Github Actions to automate Docker deployment and set up a CI/CD pipeline.
- Completed model evaluation and are refining the selected model with hyperparameter tuning.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A

Bansil Patel: N/A

Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: Faced challenges while setting up CI/Cd workflows for multiple branches in Github, particularly in aligning Docker deployment with branch specific configurations.

4.2 Your team's project blockers

- While no major blockers were encountered this week, the team experienced some delays in configuring the CI/CD pipeline for multiple branches, particularly when adjusting the Docker deployment scripts.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Investigate how GitHub Actions can be integrated into the project for automating testing and deployment.

Bansil Patel & Meet Patel: Continue analyzing the model metrics and complete the hyperparameter tuning to optimize performance.

Harsh Mohile & Rachit Bhatt: Demonstrate a fully automated Docker deployment process for multiple environments (e.g., development, staging, production) using GitHub Actions.

5.2 Your team's tasks

- Finalize the analysis of the model's performance metrics and proceed with hyperparameter optimization.
- Continue enhancing the Docker deployment workflow using GitHub Actions and ensure it works seamlessly across different environments.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

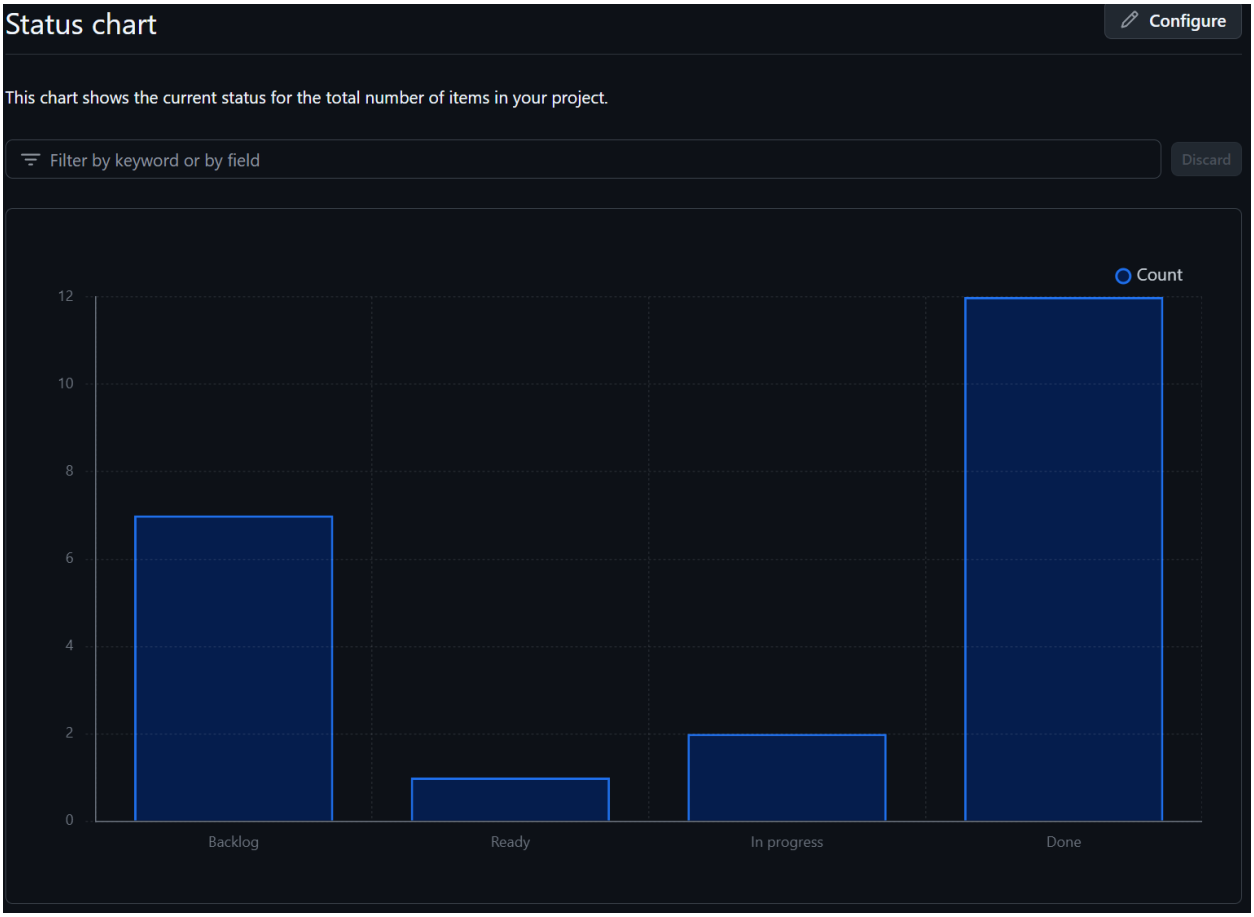

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. **Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.**

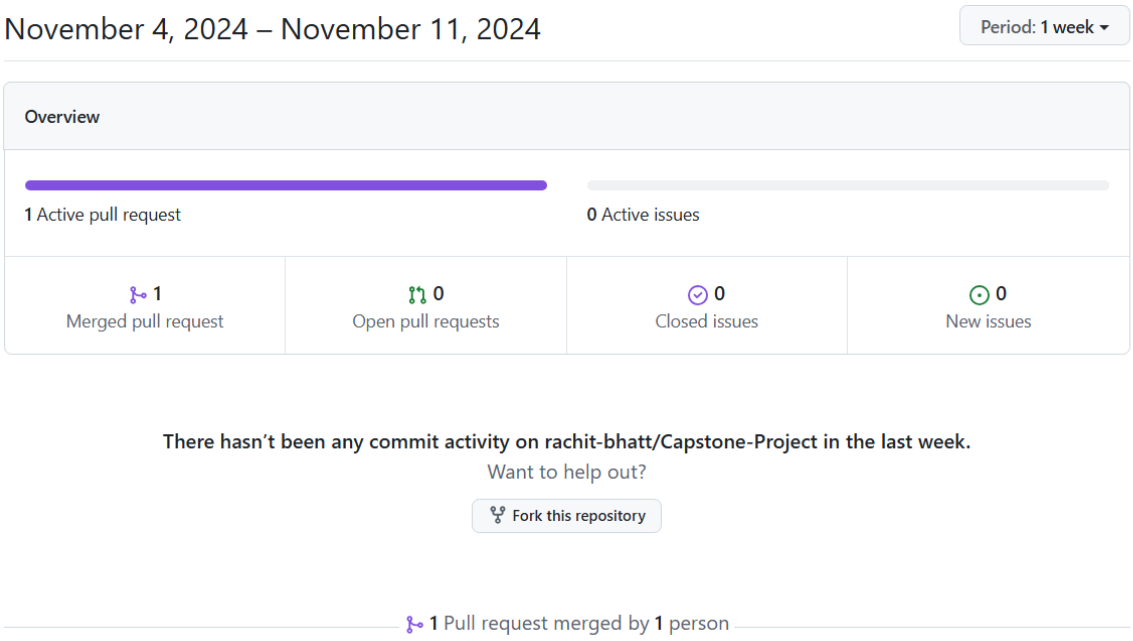

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project